RYERSON UNIVERSITY

ECN 801 PRINCIPLES OF ENGINEERING ECONOMICS

INSTRUCTOR: TSOGBADRAL GALAABAATAR

TIME: 2 HOURS

DATE: 8 FEBRUARY 2011

Instructions:

1. Choose the option closest to your answer and record the letter you have chosen on the scanner sheet. Use the space after each question to show your work. Answer all 25 questions.

2. No references (formula sheet, notes, books, etc.) are permitted.

3. Calculators that can store and display files are not allowed. Other calculators are allowed.

4. Useful formulas:

Linear Gradient Series Present Worth: $P = G\left[\frac{(1+i)^N - iN - 1}{i^2(1+i)^N}\right]$

Geometric Gradient Series Present Worth: $P = A_1\left[\frac{1-(1+g)^N(1+i)^{-N}}{i-g}\right]$ if $g \neq i$

STUDENT'S NAME:_____

(Please Print)

ID# _____

Good Luck!

1. Find the future amount of $1,000 in 5 years at 5% compounded annually.

1	**a**	1,157.20	**b**	1,300.36	**c**	1,276.30	**d**	1,250.09	**e**	5,000.00

1,000(F/P,5%,5)=1,000×1.2763=1,276.30

2. Find the present worth of future payment $2,000 six years from now at 5% compounded annually.

2	**a**	1,492.40	**b**	1,152.75	**c**	1,600.00	**d**	2,470.30	**e**	1,834.70

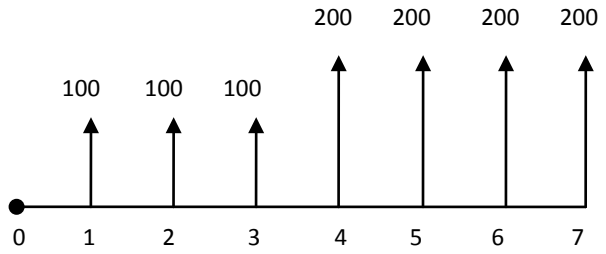
2,000(P/F,5%,6)=2,,000×0.7462=1,492.40

3. If you put $500 in a bank account and expect to have $1,500 in your account in 5 years. Find the annual interest rate of the bank.

3	**a**	31.52%	**b**	27.11%	**c**	40.5%	**d**	20.51%	**e**	24.57%

$500(1+i)^5=1,500$
$(1+i)^5=3$
$1+i=3^{1/5}$
$i=3^{1/5}-1=0.2457$

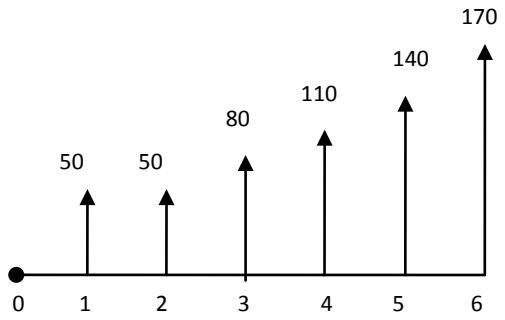
4. Find the present worth of the following cash series when interest rate is 5%.



4	**a**	1,100.00	**b**	750.45	**c**	884.94	**d**	900.52	**e**	810.71

100(P/A,5%,7)+100(P/A,5%,4)(P/F,5%,3)=100×5.7864+100×3.546×0.8636=884.94

5. Find the present worth of the following cash series when interest rate is 5%.



5	**a**	600.00	**b**	540.11	**c**	420.56	**d**	489.13	**e**	550.12

50(P/A,5%,6)+30(P/G,5%,5)(P/F,5%,1)=50×5.0757+30×8.2369×0.9524=489.13

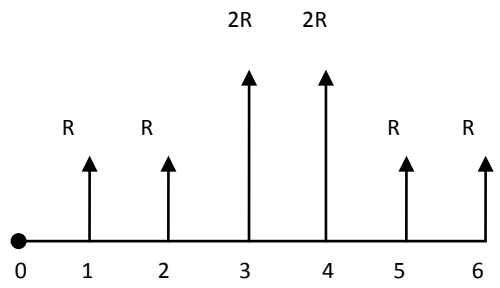
6. Suppose you borrowed $100,000 today at interest of 5% compounded annually. If you pay equal payments for 8 years starting at the end of second year, then what is the annual installment amount?

6	**a**	16,245.50	**b**	15,716.47	**c**	25,000.00	**d**	22,154.60	**e**	18,215.63

A(P/A,5%,8)(P/F,5%,1)=100,000

A×6.4632×0.9524=100.000. Thus, A=16,245.50

7. Consider the cash flow series given. Which of the following equations is incorrect?



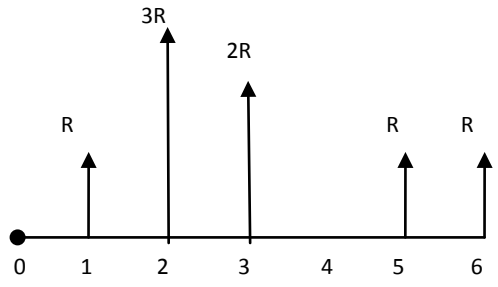
a) $P = R(P/A,i,6) + R(P/A,i,2)(P/F,i,2)$

b) $P = R(P/A,i,2) + 2R(P/A,i,2)(P/F,i,2) + R(P/A,i,2)(P/F,i,4)$

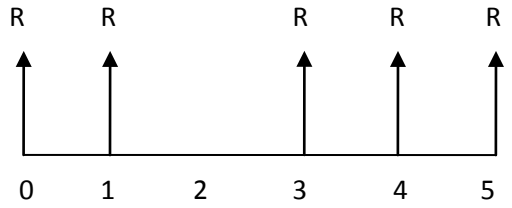
c) $F = R(F/A,i,6) + R(F/A,i,2)(F/P,i,2)$

d) $F = R(F/A,i,2)(F/P,i,4) + 2R(F/A,i,2)(F/P,i,3) + R(F/A,i,2)$

d) does not represent the given cash diagram. Instead, d) represents



8. Find the correct equation for the cash flow diagram given.



a) $P = R + R(P/A,i,5) - R(P/F,i,1)$

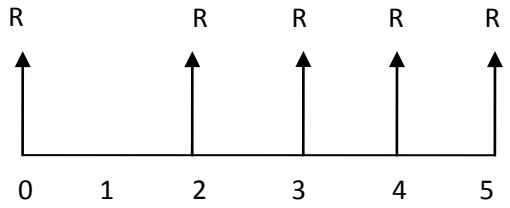
b) $P = R(P/A,i,2) + R(P/A,i,3)(P/F,i,3)$

c) $F = R(F/P,i,5) + R(F/A,i,5) - R(F/P,i,3)$

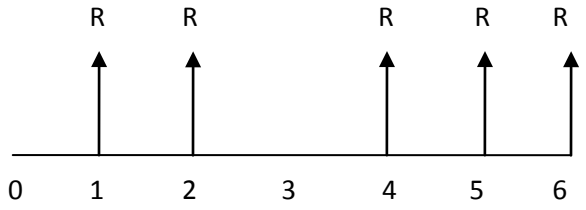
d) $F = R(F/A,i,2)(F/P,i,4) + R(F/A,i,3)(F/P,i,1)$

The correct answer is c). The cash diagrams for a), b) and d) are:

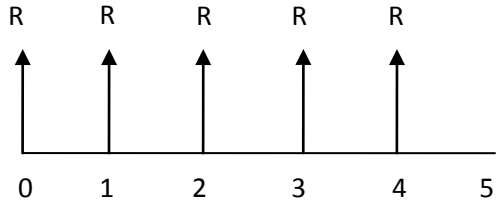
a)



b)



d)



9. According to “Rule of 72’’, how many years will it take to double your investment if the annual interest rate is 6%?

9	**a**	12	**b**	9	**c**	10	**d**	11	**e**	13

72/6=12

10. If the annual interest rate is 5%, how many (full) years you have to wait in order to triple your investment?

10	**a**	25	**b**	23	**c**	30	**d**	19	**e**	28

From the table, we find (F/P,5%,22)=2.9253 and (F/P,5%,23)=3.0715. The correct answer is b).

11. For some value of *i**, suppose *(A/G, i*,35)=7.2* and *(A/G, i*, 40)=7.389*. Find *(A/G, i*, 39)* by linear interpolation method.

11	**a**	7.312	**b**	7.306	**c**	7.297	**d**	7.351	**e**	7.374

$(A/G,i^*,39)=7.2+(7.389-7.2)\times\frac{39-35}{40-35}=7.351$

12. Which of the following equations is incorrect?

a) $(F/P,i,N)(P/F,i,N)=1$
b) $(F/P,i,N)(P/A,i,N)=(A/F,i,N)$
c) $(P/A,i,N)(A/G,i,N)=(P/G,i,N)$
d) $(F/A,i,N)=(P/A,i,N)\times(1+i)^N$

b) is incorrect. The correct version of this equation should be (F/P,i,N)(P/A,i,N)=(F/A,i,N).

13-15. Suppose you borrowed $10,000 for 4 years at interest rate of 5% compounded annually.

13. What is the equal annual payment?

13	**a**	2,820.00	**b**	2,940.00	**c**	2,775.50	**d**	2,500.00	**e**	3,000.00

A=10,000(A/P,5%,4)=10,000×0.2820=2,820

14. Find the balance outstanding at the end of second year (after the second payment).

14	**a**	5,190.00	**b**	5,087.00	**c**	5,244.00	**d**	5,301.00	**e**	4,887.00

B_2=2,820(P/A,5%,4-2)=2,820×1.8594=5,243.51≈5,244

15. Find the interest payment for the third year of loan.

15	**a**	250.50	**b**	254.30	**c**	271.60	**d**	304.70	**e**	262.20

$I_3=B_2\times0.05=262.20$

16. In the first year of production, a company is expected to sell 1,000 desktops at market price $500 per desktop. For subsequent years, the company expects 10% annual increase in quantity sold, and 5% annual increase in market desktop price. If the annual interest rate is 8%, find the present worth of first 4 years of revenue.

16	**a**	2,415,573	**b**	2,576,421	**c**	2,053,839	**d**	1,501,781	**e**	907,122

If current year's price and quantity are P and Q respectively, current year's revenue is P×Q. The next year's revenue will be 1.05P×1.1Q=1.115P×Q. In other words, revenue increases by 15.5% each year.

Now we can use the formula for geometric gradient series on the front page of the exam with g=0.155, i=0.08, N=4 and A_1=500,000 (first year's revenue).

$$P = A_1\left[\frac{1-(1+g)^N(1+i)^{-N}}{i-g}\right]$$

17. What can we say about APR compounded monthly?
    a) APR is larger than the effective annual interest rate.
    b) APR is smaller than the effective annual interest rate.
    c) APR is equal to the effective annual interest rate.

The correct answer is b).

18. If your credit card APR is 12% compounded monthly, what is the effective interest rate per quarter?

18	**a**	3.00%	**b**	3.30%	**c**	3.65%	**d**	4.25%	**e**	3.03%

Monthly effective interest rate is 12%/12=1%. There are three months in a quarter. Thus, quarterly effective interest rate is calculated by 1.01^3-1=0.030301. Thus, i=3.03%.

19. You have a choice among the following savings accounts. First one gives you 10% APR compounded monthly. The second account has 10% APR compounded daily. The last one offers 10% APR compounded quarterly. Which one of these accounts you should choose?

a) The one with monthly compounding
b) The one with daily compounding
c) The one with quarterly compounding

The correct answer is b).

20. If a loan has 20% annual interest rate compounded continuously, what is the effective interest rate per quarter? Recall that e≈2.7182.

20	**a**	5.85%	**b**	5.00%	**c**	6.54%	**d**	5.13%	**e**	6.88%

$e^{0.2/4}-1=e^{0.05}-1=0.05127\approx0.0513$. Thus the effective interest rate per quarter is 5.13%.

21. If you make quarterly deposits of $500 in savings account with 6% APR compounded monthly, how much you will have in your account in 5 years?

21	**a**	10,274	**b**	11,958	**c**	11,570	**d**	10,471	**e**	12,371

First, we find the effective quarterly rate. Since the effective monthly rate is 6%/12=0.5% , effective quarterly rate $i=1.005^3-1=0.015075$.

The total number of deposits N=5×4=20. Hence, using the formula with A=500,

$F=A\frac{(1+i)^N-1}{i}=11{,}570.37$

22. Suppose you are about to deposit your money in savings account for 2 years. You find yourself being indifferent between the following two savings accounts.
    1. 18% APR compounded semiannually.
    2. X% APR compounded monthly

    Find the value of X/12.

22	**a**	2.00%	**b**	1.50%	**c**	3.00%	**d**	2.25%	**e**	1.45%

The effective semi-annual interest rate for the first account is 9%. In order to be indifferent between them, the second account should have the same semi-annual rate.

Denote r=X/12. Then

$1.09=(1+r)^6$

r=0.0145

23. Suppose you took a loan of $5,000 with 24 monthly payments at rate of 24% APR compounded monthly. After paying 12-th payment, you want to pay off the remainder of the loan in a single payment. Compute how much you have to pay.

23	**a**	2,796	**b**	3,127	**c**	2,461	**d**	2,832	**e**	2,954

Since we can't use the table for 5% here, we need to use the corresponding formulas for following calculations.

A=5,000(A/P,2%,24)=264.5
B_{12}=264.5(P/A,4%,12)≈2,796

24. Suppose you put $5,000 in savings account. If the bank effective annual interest rate is 2% for the first 5 years and is 5% for the next 5 years, how much money will you have in your account at the end of 10-th year?

24	**a**	6,035.43	**b**	7,045.59	**c**	7,547.18	**d**	6,642.14	**e**	8,051.34

$5{,}000\times1.02^5\times1.05^5=7{,}045.59$

25. Suppose you borrowed $2,000 at add-on rate of 5% per year for four years. If you will make monthly equal payments for 48 months to pay off the loan, find the monthly payment amount.

25	**a**	55.00	**b**	60.00	**c**	40.00	**d**	65.00	**e**	50.00

Yearly interest is 2,000×0.05=100. The total interest is 100×4=400. Thus the monthly payment is (2,000+400)/48=50.

5.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor (F/P,i,N)	Present Worth Factor (P/F,i,N)	Compound Amount Factor (F/A,i,N)	Sinking Fund Factor (A/F,i,N)	Present Worth Factor (P/A,i,N)	Capital Recovery Factor (A/P,i,N)	Gradient Uniform Series (A/G,i,N)	Gradient Present Worth (P/G,i,N)	N
1	1.0500	0.9524	1.0000	1.0000	0.9524	1.0500	0.0000	0.0000	1
2	1.1025	0.9070	2.0500	0.4878	1.8594	0.5378	0.4878	0.9070	2
3	1.1576	0.8638	3.1525	0.3172	2.7232	0.3672	0.9675	2.6347	3
4	1.2155	0.8227	4.3101	0.2320	3.5460	0.2820	1.4391	5.1028	4
5	1.2763	0.7835	5.5256	0.1810	4.3295	0.2310	1.9025	8.2369	5
6	1.3401	0.7462	6.8019	0.1470	5.0757	0.1970	2.3579	11.9680	6
7	1.4071	0.7107	8.1420	0.1228	5.7864	0.1728	2.8052	16.2321	7
8	1.4775	0.6768	9.5491	0.1047	6.4632	0.1547	3.2445	20.9700	8
9	1.5513	0.6446	11.0266	0.0907	7.1078	0.1407	3.6758	26.1268	9
10	1.6289	0.6139	12.5779	0.0795	7.7217	0.1295	4.0991	31.6520	10
11	1.7103	0.5847	14.2068	0.0704	8.3064	0.1204	4.5144	37.4988	11
12	1.7959	0.5568	15.9171	0.0628	8.8633	0.1128	4.9219	43.6241	12
13	1.8856	0.5303	17.7130	0.0565	9.3936	0.1065	5.3215	49.9879	13
14	1.9799	0.5051	19.5986	0.0510	9.8986	0.1010	5.7133	56.5538	14
15	2.0789	0.4810	21.5786	0.0463	10.3797	0.0963	6.0973	63.2880	15
16	2.1829	0.4581	23.6575	0.0423	10.8378	0.0923	6.4736	70.1597	16
17	2.2920	0.4363	25.8404	0.0387	11.2741	0.0887	6.8423	77.1405	17
18	2.4066	0.4155	28.1324	0.0355	11.6896	0.0855	7.2034	84.2043	18
19	2.5270	0.3957	30.5390	0.0327	12.0853	0.0827	7.5569	91.3275	19
20	2.6533	0.3769	33.0660	0.0302	12.4622	0.0802	7.9030	98.4884	20
21	2.7860	0.3589	35.7193	0.0280	12.8212	0.0780	8.2416	105.6673	21
22	2.9253	0.3418	38.5052	0.0260	13.1630	0.0760	8.5730	112.8461	22
23	3.0715	0.3256	41.4305	0.0241	13.4886	0.0741	8.8971	120.0087	23
24	3.2251	0.3101	44.5020	0.0225	13.7986	0.0725	9.2140	127.1402	24
25	3.3864	0.2953	47.7271	0.0210	14.0939	0.0710	9.5238	134.2275	25
26	3.5557	0.2812	51.1135	0.0196	14.3752	0.0696	9.8266	141.2585	26
27	3.7335	0.2678	54.6691	0.0183	14.6430	0.0683	10.1224	148.2226	27
28	3.9201	0.2551	58.4026	0.0171	14.8981	0.0671	10.4114	155.1101	28
29	4.1161	0.2429	62.3227	0.0160	15.1411	0.0660	10.6936	161.9126	29
30	4.3219	0.2314	66.4388	0.0151	15.3725	0.0651	10.9691	168.6226	30
31	4.5380	0.2204	70.7608	0.0141	15.5928	0.0641	11.2381	175.2333	31
32	4.7649	0.2099	75.2988	0.0133	15.8027	0.0633	11.5005	181.7392	32
33	5.0032	0.1999	80.0638	0.0125	16.0025	0.0625	11.7566	188.1351	33
34	5.2533	0.1904	85.0670	0.0118	16.1929	0.0618	12.0063	194.4168	34
35	5.5160	0.1813	90.3203	0.0111	16.3742	0.0611	12.2498	200.5807	35
40	7.0400	0.1420	120.7998	0.0083	17.1591	0.0583	13.3775	229.5452	40
45	8.9850	0.1113	159.7002	0.0063	17.7741	0.0563	14.3644	255.3145	45
50	11.4674	0.0872	209.3480	0.0048	18.2559	0.0548	15.2233	277.9148	50
55	14.6356	0.0683	272.7126	0.0037	18.6335	0.0537	15.9664	297.5104	55
60	18.6792	0.0535	353.5837	0.0028	18.9293	0.0528	16.6062	314.3432	60
65	23.8399	0.0419	456.7980	0.0022	19.1611	0.0522	17.1541	328.6910	65
70	30.4264	0.0329	588.5285	0.0017	19.3427	0.0517	17.6212	340.8409	70
75	38.8327	0.0258	756.6537	0.0013	19.4850	0.0513	18.0176	351.0721	75
80	49.5614	0.0202	971.2288	0.0010	19.5965	0.0510	18.3526	359.6460	80
85	63.2544	0.0158	1245.0871	0.0008	19.6838	0.0508	18.6346	366.8007	85
90	80.7304	0.0124	1594.6073	0.0006	19.7523	0.0506	18.8712	372.7488	90
95	103.0347	0.0097	2040.6935	0.0005	19.8059	0.0505	19.0689	377.6774	95
100	131.5013	0.0076	2610.0252	0.0004	19.8479	0.0504	19.2337	381.7492	100

© CourseSmart